Hierarchy of Criteria

1 Illustrative example: Choosing a location for a fish restaurant

An entrepreneur plans to open a fish restaurant in one of four locations: Gdańsk, Rozewie, Słupsk, or Elbląg. The entrepreneur considers five criteria:

K_1 Expected annual demand (thousands of customers): **maximize**.

K_2 Annual operating costs, including transport, rent, and utilities (thousands of PLN): **minimize**.

K_3 Number of competing fish restaurants: **minimize**.

K_4 Availability of qualified employees (scale 1–10): **maximize**.

K_5 Operational flexibility (scale 1–10): **maximize**.

Operational flexibility measures the ability to adjust opening hours, staffing, and operating capacity in response to fluctuations in demand, including seasonal changes. It does not measure the operating costs already included in K_2.

Table 1: Fictional data for an illustrative multi-criteria location problem

Location	K_1	K_2	K_3	K_4	K_5
Gdańsk	85	420	32	9	6
Rozewie	38	230	18	3	10
Słupsk	36	265	9	7	7
Elbląg	43	280	12	8	7
Direction	Max	Min	Min	Max	Max

Disclaimer. This example is intended solely for educational purposes. All numerical values are entirely fictional and do not represent actual economic, demographic, or market conditions in any of the locations considered. The example is not intended to promote or discourage investment in any city or locality, nor to assess its actual attractiveness for business.

With rows ordered as in Table 1, the decision matrix is

$$Q = \begin{bmatrix} 85 & 420 & 32 & 9 & 6 \\ 38 & 230 & 18 & 3 & 10 \\ 36 & 265 & 9 & 7 & 7 \\ 43 & 280 & 12 & 8 & 7 \end{bmatrix}.$$

The criteria K_1, K_4, and K_5 are *stimulants*, whereas K_2 and K_3 are *destimulants*.

Exercises. Transform the destimulants into stimulants and normalize the decision matrix. Identify the Pareto-efficient alternatives. Then assign weights to the five criteria and compare the locations using a weighted-sum method. Discuss how the ranking changes as the weight assigned to operational flexibility increases.

2 Formulation of the model

2.1 Stimulants and destimulants

Let A be a nonempty set of decision alternatives, and let $f_j : A \to \mathbb{R}_{++} = (0, \infty)$ denote the value of criterion j. Let $\succ_j$ denote strict preference with respect to this criterion.

Definition 1 (Stimulant (benefit criterion))**.** *A criterion f_j is called a **stimulant** if larger values are strictly preferred to smaller ones. Formally, for all $a, b \in A$,*

$$a \succ_j b \quad \Longleftrightarrow \quad f_j(a) > f_j(b).$$

Definition 2 (Destimulant (cost criterion))**.** *A criterion f_j is called a **destimulant** if*

smaller values are strictly preferred to larger ones. Formally, for all $a, b \in A$,

$$a \succ_j b \quad \iff \quad f_j(a) < f_j(b).$$

Thus, a stimulant is to be maximized, whereas a destimulant is to be minimized. Equal criterion values correspond to neither alternative being strictly preferred under that criterion.

Example 1. *In the fish-restaurant location problem, expected demand* (K_1)*, availability of employees* (K_4)*, and operational flexibility* (K_5) *are stimulants. Operating costs* (K_2) *and the number of competitors* (K_3) *are destimulants.*

Example 2. *In the Markowitz portfolio-selection problem, we consider two criteria for a portfolio: its expected rate of return (a stimulant) and its variance (a destimulant).*

2.2 Normalization

Assume that A is finite and write $A = \{a_1, \ldots, a_n\}$. We consider m criteria $f_1, \ldots, f_m$ with strictly positive values. First, transform each destimulant into a stimulant by taking its reciprocal:

$$g_j(a_i) = \begin{cases} f_j(a_i), & \text{if } f_j \text{ is a stimulant,} \\ \dfrac{1}{f_j(a_i)}, & \text{if } f_j \text{ is a destimulant.} \end{cases}$$

Since all values are positive, this transformation preserves the preferred direction: larger values of g_j are always better. Next, normalize each criterion *across the alternatives*:

$$\boxed{h_j(a_i) = \frac{g_j(a_i)}{\sum_{r=1}^{n} g_j(a_r)}, \qquad i = 1, \ldots, n, \quad j = 1, \ldots, m.}$$

Thus $0 < h_j(a_i) < 1$ for $n > 1$, and

$$\sum_{i=1}^{n} h_j(a_i) = 1 \qquad \text{for every } j.$$

Every normalized criterion is a stimulant. Note that these are *column sums*; in general, the values in a row do not sum to one. The best alternative in a criterion need not receive the value 1, and the worst need not receive the value 0.

Example 3 (Normalization of the fish-restaurant data)**.** *Consider the matrix Q in Table 1. For the stimulant K_1 (expected demand), the column sum equals* $85+38+36+43 = 202$*, and hence*

$$h_1(\textit{Rozewie}) = \frac{38}{202} \approx 0.1881.$$

For the destimulant K_2 (operating costs), we first take reciprocals:

$$g_2(\textit{Gdańsk}) = \frac{1}{420}, \quad g_2(\textit{Rozewie}) = \frac{1}{230}, \quad g_2(\textit{Słupsk}) = \frac{1}{265}, \quad g_2(\textit{Elbląg}) = \frac{1}{280}.$$

Thus the normalized value for Słupsk is

$$h_2(\textit{Słupsk}) = \frac{1/265}{1/420 + 1/230 + 1/265 + 1/280} \approx 0.2681.$$

Applying the same procedure to every column yields

$$H = \begin{bmatrix} 0.4208 & 0.1692 & 0.1111 & 0.3333 & 0.2000 \\ 0.1881 & 0.3089 & 0.1975 & 0.1111 & 0.3333 \\ 0.1782 & 0.2681 & 0.3951 & 0.2593 & 0.2333 \\ 0.2129 & 0.2538 & 0.2963 & 0.2963 & 0.2333 \end{bmatrix}.$$

All entries are rounded to four decimal places. Rows are ordered as Gdańsk, Rozewie, Słupsk, and Elbląg, and columns as $K_1, \ldots, K_5$.

The transformations preserve Pareto dominance. For these fictional data, no location dominates another: all four are Pareto-efficient.

2.3 Gibbs entropy

The concept of entropy originates from thermodynamics and statistical mechanics. It provides a mathematical measure of uncertainty or dispersion in a probability distribution.

Table 2: Column-normalized fish-restaurant data: all criteria are stimulants

Location	K_1	K_2	K_3	K_4	K_5
Gdańsk	0.4208	0.1692	0.1111	0.3333	0.2000
Rozewie	0.1881	0.3089	0.1975	0.1111	0.3333
Słupsk	0.1782	0.2681	0.3951	0.2593	0.2333
Elbląg	0.2129	0.2538	0.2963	0.2963	0.2333
Direction	Max	Max	Max	Max	Max

Consider a finite set of possible states $\Omega = \{\omega_1, \dots, \omega_n\}$, with probabilities $p_1, \dots, p_n$ satisfying

$$p_i \geq 0, \qquad \sum_{i=1}^{n} p_i = 1.$$

The Gibbs entropy is defined by

$$S(p) = -k_B \sum_{i=1}^{n} p_i \ln p_i,$$

where $k_B > 0$ is the Boltzmann constant. We adopt the convention $0 \ln 0 = 0$.

In information theory, the corresponding dimensionless quantity is

$$H(p) = -\sum_{i=1}^{n} p_i \ln p_i.$$

Proposition 1. *For every probability distribution $p = (p_1, \dots, p_n)$,*

$$0 \leq H(p) \leq \ln n.$$

Moreover, $H(p) = 0$ if and only if $p = e^i$ for some $i \in \{1, \dots, n\}$, where e^i is the i-th unit vector; and $H(p) = \ln n$ if and only if $p_i = 1/n$ for every i.

Proof. Let $\phi(0) = 0$ and $\phi(t) = t \ln t$ for $t > 0$. Since $\lim_{t \to 0^+} t \ln t = 0$, the function ϕ is continuous on $[0, 1]$. Moreover, $\phi''(t) = 1/t > 0$ for $t > 0$, so ϕ is strictly convex. By Jensen's inequality,

$$\frac{1}{n} \sum_{i=1}^{n} \phi(p_i) \geq \phi\left(\frac{1}{n} \sum_{i=1}^{n} p_i \right) = \phi\left(\frac{1}{n} \right) = -\frac{\ln n}{n}.$$

Multiplying by $-n$ yields $H(p) \leq \ln n$. Strict convexity shows that equality holds precisely for the uniform distribution. On the other hand, $-p_i \ln p_i \geq 0$ for every $p_i \in [0, 1]$. Hence $H(p) \geq 0$, with equality if and only if each $p_i \in \{0, 1\}$. Since the probabilities sum to one, this is equivalent to $p = e^i$ for some i. □

Thus entropy is maximal for a uniform distribution and minimal for a distribution concentrated on a single state.

Example 4 (Entropy for four possible states)**.** *Consider the same number of states $n = 4$ in the following three distributions:*

$$p = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right), \qquad q = (1, 0, 0, 0), \qquad r = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0\right).$$

Direct calculation gives

$$H(p) = \ln 4, \qquad H(q) = 0, \qquad H(r) = \frac{3}{2} \ln 2.$$

The uniform distribution p has maximal entropy; q, concentrated on one state, has minimal entropy; and r is an intermediate case. In the fish-restaurant problem there are also four alternatives, but the entropy-based criterion weights use normalized criterion values*, not probabilities of the four locations being chosen.*

2.4 Entropy and criterion weights

Entropy can also be used to determine objective weights in multi-criteria decision analysis.

The basic idea is that a criterion which assigns similar evaluations to all alternatives provides little information for distinguishing them.

Conversely, a criterion whose evaluations are strongly differentiated may receive a higher weight.

Suppose that $h_j(a_i) \in [0, 1]$ are the normalized evaluations of n alternatives according to criterion K_j, and all criteria are stimulants.

For each criterion with $\sum_{i=1}^{n} h_j(a_i) > 0$, define

$$p_{ij} = \frac{h_j(a_i)}{\sum_{r=1}^{n} h_j(a_r)}.$$

Thus,

$$p_{ij} \geq 0, \qquad \sum_{i=1}^{n} p_{ij} = 1.$$

The normalized entropy of criterion K_j is

$$E_j = -\frac{1}{\ln n} \sum_{i=1}^{n} p_{ij} \ln p_{ij}.$$

Consequently, $0 \leq E_j \leq 1$.

Define the degree of differentiation by

$$D_j = 1 - E_j.$$

Finally, the entropy weight of criterion K_j is

$$\boxed{w_j = \frac{D_j}{\sum_{k=1}^{m} D_k} = \frac{1 - E_j}{\sum_{k=1}^{m} (1 - E_k)}.}$$

We assume that the denominator is positive. A criterion for which all normalized evaluations are zero is treated as non-discriminating and assigned $D_j = 0$.

The weights satisfy

$$w_j \geq 0, \qquad \sum_{j=1}^{m} w_j = 1.$$

Remark 1. *Entropy weights measure the ability of criteria to distinguish alternatives. They do not necessarily represent the subjective importance of criteria to the decision-maker.*

2.5 The original Simos method

The method proposed by Jean Simos derives subjective criterion weights from an arrangement of cards. The decision-maker orders the criteria from least to most important,

groups equally important criteria, and may insert blank cards between successive levels to indicate larger differences in importance.
The decision-maker follows the algorithm below.

1. Write the name of each criterion on a separate card.
2. Rank the criteria from least to most important. Cards representing equally important criteria are grouped together.
3. Arrange the cards and groups of cards in a stack, placing the least important criteria at the bottom and the most important at the top.
4. Insert blank cards between successive groups to indicate differences in importance. More blank cards indicate a larger difference.
5. Number all cards consecutively from bottom to top, including the blank cards.
6. Assign each criterion the number of its card. If several criteria belong to the same group, assign each of them the arithmetic mean of the numbers occupied by the cards in that group.
7. Normalize the assigned values by dividing each by the sum of the values assigned to all criteria. The resulting weights sum to one.

Example 5 (The original Simos method)**.** *Consider the five criteria in the fish-restaurant example. Suppose the entrepreneur gives the following ranking, from least to most important:*

$$K_5 \mid \square \mid \{K_3, K_4\} \mid \square\ \square \mid K_1 \mid K_2.$$

Here $\square$ *denotes a blank card, and*

$L_1 = \{K_5\}, \quad L_2 = \{K_3, K_4\}, \quad L_3 = \{K_1\}, \quad L_4 = \{K_2\},$

$(n_1, n_2, n_3, n_4) = (1, 2, 1, 1)$, *sizes of groups*

$(e'_1, e'_2, e'_3) = (1, 2, 0)$ *count of blank cards between following criteria or groups.*

The consecutive positions give

Level	*Occupied positions*	s_ℓ
L_1	1	1
L_2	$3, 4$	$7/2$
L_3	7	7
L_4	8	8

The denominator is $\sum_{\ell=1}^{4} n_\ell s_\ell = 1 + 2(7/2) + 7 + 8 = 23$. *In the order* $(K_1, \ldots, K_5)$,

$$\boxed{w^S = \left(\frac{7}{23}, \frac{8}{23}, \frac{7}{46}, \frac{7}{46}, \frac{1}{23}\right).}$$

The resulting largest-to-smallest weight ratio is 8.

Why was the original procedure revised? In the original method the ratio of the greatest to the smallest weight is imposed by the numbering of the cards. The decision-maker has not been asked whether the most important criterion should be exactly eight times as important as the least important criterion. Furthermore, the scores depend on the number of cards within tied levels; this can change the interpretation of the gaps. The revised procedure of Figueira and Roy [1] separates the ranking and the spacing information from an explicitly elicited largest-to-smallest weight ratio.

2.6 The revised Simos method (Figueira–Roy)

Let us denote importance levels of groups of cards $L_1, \ldots, L_r$ ordered from the least important to the most important, their sizes n_ℓ, and the blank-card counts e'_ℓ. The decision-maker now specifies

$$s = \frac{\text{weight of the most important criterion}}{\text{weight of the least important criterion}} > 1.$$

Define the number of intervals between successive levels by

$$e_\ell = e'_\ell + 1, \qquad e = \sum_{\ell=1}^{r-1} e_\ell, \qquad u = \frac{s-1}{e}.$$

The unnormalized common weight at level L_ℓ is

$$\boxed{v_\ell = 1 + u \sum_{h=1}^{\ell-1} e_h, \qquad \ell = 1, \ldots, r.}$$

Hence $v_1 = 1$ and $v_r = z$. For each $K_j \in L_\ell$,

$$\boxed{w_j^{RS} = \frac{v_\ell}{\sum_{h=1}^{r} n_h v_h}.}$$

This formula gives the unrounded weights. The complete revised procedure also specifies rounding rules for prescribed weight precision; here we retain exact weights and round only displayed values.

Example 6 (The revised weights for the same five criteria)**.** *Use exactly the same ranking and blank cards as before:*

$$(n_1, n_2, n_3, n_4) = (1, 2, 1, 1), \qquad (e'_1, e'_2, e'_3) = (1, 2, 0).$$

Then $(e_1, e_2, e_3) = (2, 3, 1)$ *and* $e = 6$. *Suppose the entrepreneur specifies* $s = 4$. *We obtain*

$$u = \frac{4-1}{6} = \frac{1}{2}, \qquad (v_1, v_2, v_3, v_4) = \left(1, 2, \frac{7}{2}, 4\right).$$

As $\sum_{\ell=1}^{4} n_\ell v_\ell = 1 + 2(2) + 7/2 + 4 = 25/2$,

$$\boxed{w^{RS} = \left(\frac{7}{25}, \frac{8}{25}, \frac{4}{25}, \frac{4}{25}, \frac{2}{25}\right) = (0.28, 0.32, 0.16, 0.16, 0.08).}$$

Thus $w_2^{RS}/w_5^{RS} = 4$ *as explicitly specified.*

Remark 2. *Entropy and Simos weights represent different ideas. Entropy measures how much a criterion differentiates the alternatives under a chosen numerical normalization.*

The Simos procedures elicit the decision- maker's relative judgements of criterion importance. Neither method by itself measures the investor's risk preferences in expected-utility terms. Weighted sums also depend on the scales of the normalized criterion values.

2.7 Illustration: entropy weights versus Simos weights in a brokerage office

The following fictional example illustrates the difference between *objective* and *subjective* weighting methods.

A very risk-averse investor visits a brokerage office and wants to invest a small amount of money, say PLN 100, in one of four available assets. The investor is mainly afraid of risk and initially wants to choose the least risky asset.

The broker, however, points out that the variances of the available assets are very similar. Hence, from the broker's perspective, risk hardly distinguishes the alternatives, whereas expected return does. The broker therefore suggests that, for such a small trial investment, the investor may focus more on profitability.

We consider two criteria:

- K_1 – expected annual return (stimulant);
- K_2 – variance (destimulant).

Table 3: Fictional data for four financial assets

Asset	K_1: expected return (%)	K_2: variance
A	5	0.0390
B	8	0.0400
C	11	0.0410
D	14	0.0405
Direction	Max	Min

Comment. The data are entirely fictional and are used only for didactic purposes. The amount of PLN 100 is chosen only to emphasize that the investor treats this as a small trial investment rather than a major financial decision.

Entropy interpretation

Use the same transformation and column normalization as in the Normalization subsection:

$$g_1(a_i) = f_1(a_i), \qquad g_2(a_i) = \frac{1}{f_2(a_i)}, \qquad h_j(a_i) = \frac{g_j(a_i)}{\sum_{r=1}^{4} g_j(a_r)}.$$

The transformed criterion vectors are

$$g_1 = (5, 8, 11, 14), \qquad g_2 \approx (25.6410, 25.0000, 24.3902, 24.6914).$$

Therefore, the column-normalized matrix is

$$\boxed{H = \begin{bmatrix} 0.131579 & 0.257123 \\ 0.210526 & 0.250695 \\ 0.289474 & 0.244581 \\ 0.368421 & 0.247600 \end{bmatrix}}.$$

Since every column sums to one, the probabilities used for entropy are simply $p_{ij} = h_j(a_i)$: no further normalization is needed. For $n = 4$ we obtain

$$E_j = -\frac{1}{\ln 4} \sum_{i=1}^{4} h_j(a_i) \ln h_j(a_i),$$

with

$$(E_1, E_2) \approx (0.953354, 0.999876), \qquad (D_1, D_2) = (1 - E_1, 1 - E_2).$$

Thus, the entropy weights are

$$\boxed{w^{\text{ent}} \approx (0.997344, 0.002656).}$$

The variance column is nearly uniform, so the entropy procedure assigns it very little discrimination weight.

Simos interpretation

The investor instead considers risk more important than expected return. The ordering from *least* to *most* important is $K_1 \mid \square \mid K_2$, i.e.

$$L_1 = \{K_1\}, \quad L_2 = \{K_2\}, \quad (n_1, n_2) = (1, 1), \quad e_1' = 1.$$

The original Simos procedure assigns positions 1 and 3, so $s_1 = 1$, $s_2 = 3$, and

$$\boxed{w^S = \left(\frac{1}{4}, \frac{3}{4}\right).}$$

For completeness, the revised procedure would use $e_1 = e_1' + 1 = 2$ and a ratio z explicitly chosen by the investor. If $z = 3$, then $u = (3-1)/2 = 1$ and $w^{RS} = (1/4, 3/4)$; other choices of z yield different weights.

Ranking the same four assets

To compare the two weighting rules consistently, use the *same* column-normalized matrix H above. With weights w, the weighted score of asset a_i is

$$S(a_i) = \sum_{j=1}^{2} w_j h_j(a_i).$$

Under the original Simos weights,

Asset	A	B	C	D
S^S	0.225737	0.240653	0.255804	0.277806

so asset D ranks first. Under entropy weights,

Asset	A	B	C	D
S^{ent}	0.131912	0.210633	0.289354	0.368100

and asset D also ranks first, though expected return is far more heavily weighted.

Remark 3. *Even though the original Simos procedure assigns* 75% *of the weight to variance, the numerical variance differences in the column-normalized matrix are so small that the weighted sum still selects the asset with the highest expected return. Therefore a high subjective weight for risk does not, by itself, guarantee the selection of the least-risky asset. The result depends both on the weights and on the chosen normalization. In particular, one must not mix column-sum normalization in the entropy calculation with min–max normalization in the final weighted scores when the goal is a comparison under a single normalization rule.*

Moreover, the hypothetical loss of PLN 100 does not justify changing an investor's risk preferences: an appropriate decision should also consider the investor's financial circumstances and potential losses, not only the dispersion of the criterion table.

References

[1] Figueira, J., and Roy, B. (2002). Determining the weights of criteria in the ELECTRE type methods with a revised Simos' procedure. *European Journal of Operational Research*, 139(2), 317–326.

[2] Kobryń, A. (2014). Wielokryterialne wspomaganie decyzji w gospodarowaniu przestrzennym.